BONUS ASSIGNMENT: EGOS OPERATING SYSTEM

Contributors: 1) Aryan Kalpesh Pancholi (2024122)
2) Garv Jain (2024216)

Contributions Part 1:

Contribution by Aryan Kalpesh Pancholi (Roll No. 2024122)

- Contribution in grep.c:
    - Implemented the logic for matching substring patterns.
    - Checked for pattern edge cases.
    - Contributed to line-buffer management.
    - Ensured that the argument is valid and handled errors.
    - Ensured that it prints only the matched lines.
- Contribution in wcl.c:
    - Implemented the code for counting total lines.
    - Looked for the helper function.
    - Helped in fixing corner cases and tested wcl.

Contribution by Garv Jain (Roll No. 2024216)

- Contribution in grep.c:
    - Implemented the code for parsing the arguments and for validating argc.
    - Implemented the code for the extraction of lines from the file buffer
    - Ensured that line-buffer doesn’t overflow

    - Supported in ensuring that it prints only the matched lines.
- Contribution in wcl.c:
    - Implemented the code for reading multiple files.
    - Handled errors and missing file cases.
    - Implemented line-counting helper in the main loop.

Contributions Part 2:

Contribution by Aryan Kalpesh Pancholi (Roll No. 2024122):

- Initialized the lifecycle and MLFQ-related fields/variables.
- Implemented the logic for runtime-based priority demotion in the MLFQ update part.
- Implemented the PCB struct in the process.h file required for MLFQ scheduling.
- Helped in implementing the lifecycle statistic update when a process terminates.
- Implemented code for response time and last schedule time.

Contribution by Garv Jain (Roll No. 2024216):

- Implemented the MLFQ-based scheduling logic.
- Implemented the code for handling the timer interrupt for scheduling.
- Implemented the logic in proc_yield() to handle the remaining system call processes.
- Implemented the code for MLFQ periodic reset
- Looked after the accounting of the process time-usage .

Git Repo Link - Bonus_Repo